CHAPTER 8

Prime numbers and Fermat's little theorem

Infinitude of the primes, factorisation, Fermat's little theorem

Objectives. Recognise a prime number; prove that there are infinitely many primes; factorise an integer into primes and use the factorisation (divisors, gcd); use Fermat's little theorem to compute remainders.

Prerequisites. Chapters 6 and 7; binomial coefficients; binomial formula (chapter 1).

Fluency

Fluency · No calculator, five minutes.

1. Is the integer 91 prime?
2. Give the prime numbers less than 30.
3. Factorise 360 into primes.
4. Factorise 2,026 into primes, given that 1,013 is prime.
5. Compute 2^{10} and give its remainder in the division by 11.
6. Are the integers 35 and 72 coprime?